Four Monoidal Structures on Containers: Coproduct, Product, and Extension Comparisons

Extracted from a machine-checked development

Abstract

The category **Cont** of containers carries (at least) four monoidal structures of interest: the coproduct $+$, the categorical product $\times$, the Dirichlet tensor $\otimes$, and the sequential operator $\triangleleft$. We complete the account of the first two: bifunctoriality, structural isomorphisms, naturality, and the pentagon and triangle coherence laws. We further prove the comparison isomorphisms that make the extension functor $\mathrm{Ext}(-)$ strong monoidal for the coproduct and the product,

$$\mathrm{Ext}(C + D)\, X \;\cong\; \mathrm{Ext}\, C\, X \;\oplus\; \mathrm{Ext}\, D\, X,$$
$$\mathrm{Ext}(C \times D)\, X \;\cong\; \mathrm{Ext}\, C\, X \;\times\; \mathrm{Ext}\, D\, X,$$

together with the unit comparisons and the shared identity comparison $\mathrm{Ext}\, y\, X \cong X$. The product/Dirichlet contrast is emphasised: product and Dirichlet share shapes $S \times T$ and differ only in positions ($P(s) \oplus Q(t)$ versus $P(s) \times Q(t)$), yet the extension of the product is the pointwise product of functors by the exponential law $X^{A+B} \cong X^A \times X^B$. The proofs of all stated results have been machine-checked; we record only the mathematical content.

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1 Introduction

Containers present polynomial endofunctors. The category **Cont** of containers and their morphisms admits several tensors, each with a different reading on extensions:

structure	tensor of (S, P), (T, Q)	unit	Ext pointwise?
coproduct $+$	$(S \oplus T,\ \mathsf{elim}(P, Q))$	0	yes
product $\times$	$(S \times T,\ (s, t) \mapsto P(s) \oplus Q(t))$	1	yes
Dirichlet $\otimes$	$(S \times T,\ (s, t) \mapsto P(s) \times Q(t))$	y	no
sequential $\triangleleft$	$(\sum_s (P(s) \to T),\ \ldots)$	$I{=}y$	no

The sequential and Dirichlet structures are treated in companion notes. Here we supply the monoidal data for $+$ and $\times$, and the extension comparisons that make $\mathrm{Ext}(-)$ strong monoidal for both.

The product/Dirichlet contrast. Product and Dirichlet have the same shapes $S \times T$. They differ only in positions: a sum versus a product. Yet the *extensions* invert the reading. The container whose positions are a sum has extension the pointwise product of functors, because a labelling of $P(s) \oplus Q(t)$ *is* a pair of labellings. The Dirichlet positions $P(s) \times Q(t)$ admit no such splitting, and $\mathrm{Ext}(C \otimes D)$ is not any pointwise combination of $\mathrm{Ext}\, C$ and $\mathrm{Ext}\, D$.

What this note claims.

1. $(\mathbf{Cont}, +, 0)$ is monoidal.
2. $(\mathbf{Cont}, \times, 1)$ is monoidal.
3. $\mathrm{Ext}(C + D)\, X \cong \mathrm{Ext}\, C\, X \oplus \mathrm{Ext}\, D\, X$ and $\mathrm{Ext}\, 0\, X \cong \emptyset$.
4. $\mathrm{Ext}(C \times D)\, X \cong \mathrm{Ext}\, C\, X \times \mathrm{Ext}\, D\, X$ and $\mathrm{Ext}\, 1\, X \cong 1$.
5. $\mathrm{Ext}\, y\, X \cong X$ (shared unit comparison for $\otimes$ and $\triangleleft$).

None of the comparison maps is an identity: each is a genuine isomorphism, which is exactly what "strong, not strict" means.

We work throughout in a single universe of types. Proofs are indicated by the phrase "by lean."

2 Background and units

Definition 2.1 (Container and extension)**.** A **container** $C = (S, P)$ consists of a shape type S and a position family P. Its **extension** at a type X is

$$\mathrm{Ext}\, C\, X \;=\; \sum_{s\,:\,S} \big(P(s) \to X\big). \tag{1}$$

Morphisms of containers are shape maps with contravariant position maps; we write **Cont** for the category.

Definition 2.2 (Empty and unit containers)**.**

1. The **empty container** 0 has shape type $\emptyset$ (and, vacuously, no positions). Its extension is the constantly empty functor. It is the unit for the coproduct and the initial object of **Cont**.
2. The **one-shape, no-position container** 1 has shape type 1 (a singleton) and empty position set. Its extension is the constantly-one functor—the polynomial y^0. It is the unit for the categorical product and the terminal object of **Cont**.

Note that $1 = y^0$ and $y = y^1$: the former is the unit for $\times$, the latter for $\otimes$ and $\triangleleft$.

We take as given the coproduct $C + D$ and product $C \times D$ of containers (with their universal properties): shapes of the coproduct are $S \oplus T$ with positions $\mathsf{elim}(P, Q)$; shapes of the product are $S \times T$ with positions $P(s) \oplus Q(t)$.

3 The coproduct monoidal structure

Definition 3.1 (Action on morphisms)**.** Given morphisms $\mu : C \to C'$ and $\nu : D \to D'$, the **coproduct action**

$$\mu +_2 \nu \;:\; C + D \;\longrightarrow\; C' + D'$$

is obtained from the coproduct's universal property: it is the copairing of μ then the left injection and ν then the right injection.

Proposition 3.2 (Functor laws for $+$)**.** *1.* $\mathrm{id}_C +_2 \mathrm{id}_D = \mathrm{id}_{C+D}$*;*

2. $(\mu\,;\mu') +_2 (\nu\,;\nu') = (\mu +_2 \nu)\,;(\mu' +_2 \nu')$.

Proof. By lean. □

Definition 3.3 (Structural isomorphisms for $+$)**.** 1. The **associator** $(C + D) + E \xrightarrow{\sim} C + (D + E)$ is $\oplus$-reassociation on shapes and the identity on positions in every branch (the positions of an injected shape *are* the positions of the summand).

2. The **left unitor** $0 + C \xrightarrow{\sim} C$ and **right unitor** $C + 0 \xrightarrow{\sim} C$ collapse the empty summand.

Proposition 3.4 (Naturality for $+$)**.** *The associator and both unitors of the coproduct are natural in all arguments.*

Proof. By lean. □

Theorem 3.5 (Coherence for $+$)**.** *The pentagon and triangle identities hold for the coproduct tensor.*

Proof. By lean. □

Theorem 3.6 ((**Cont**$, +, 0$) is monoidal)**.** *The category of containers, equipped with the coproduct as tensor and the empty container as unit, is a monoidal category.*

Proof. By lean. □

Remark 3.7. In a formal development that admits only one canonical monoidal instance on **Cont**, the sequential structure is preferred and the coproduct is packaged as named data rather than a global instance. Mathematically there is no conflict: four monoidal structures on one category is the point.

4 The product monoidal structure

Definition 4.1 (Action on morphisms)**.** Given morphisms $\mu : C \to C'$ and $\nu : D \to D'$, the **product action**

$$\mu \times_2 \nu \;:\; C \times D \;\longrightarrow\; C' \times D'$$

is obtained from the product's universal property: it is the pairing of the first projection then μ with the second projection then ν.

Proposition 4.2 (Functor laws for $\times$)**.** *1.* $\mathrm{id}_C \times_2 \mathrm{id}_D = \mathrm{id}_{C\times D}$*;*

2. $(\mu\,;\mu') \times_2 (\nu\,;\nu') = (\mu \times_2 \nu)\,;(\mu' \times_2 \nu')$.

Proof. By lean. □

Definition 4.3 (Structural isomorphisms for $\times$)**.** 1. The **associator** $(C \times D) \times E \xrightarrow{\sim} C \times (D \times E)$ is product reassociation on shapes and (backward) sum reassociation on positions: positions of the product are sums, so the associator must reassociate those sums in the opposite direction from the shapes.

2. The **left unitor** $1 \times C \xrightarrow{\sim} C$ and **right unitor** $C \times 1 \xrightarrow{\sim} C$: the unit shape contributes no positions, so positions of the product are just those of C (injected into the appropriate summand).

Proposition 4.4 (Naturality for $\times$)**.** *The associator and both unitors of the product are natural in all arguments.*

Proof. By lean. □

Theorem 4.5 (Coherence for $\times$)**.** *The pentagon and triangle identities hold for the product tensor.*

Proof. By lean. □

Theorem 4.6 ((**Cont**, $\times$, 1) is monoidal)**.** *The category of containers, equipped with the categorical product as tensor and the one-shape no-position container as unit, is a monoidal category.*

Proof. By lean. □

Remark 4.7. Shapes multiply, but positions add. That is the subtlety of the product tensor, and the reason its extension comparison (Section 5) is the exponential law rather than a mere unpacking of definitions.

5 Extension as a monoidal functor

For a monoidal functor one needs, for each monoidal structure on the source, a tensor on the target (endofunctors of **Type**) and comparison maps. Here:

- $+$ maps to the pointwise coproduct of functors, 0 to the constantly empty functor;
- $\times$ maps to the pointwise product of functors, 1 to the constantly-one functor;
- $\triangleleft$ maps to composition of functors (companion note);
- $\otimes$ admits *no* pointwise tensor on the target (companion note on the Dirichlet structure).

Each comparison below is a genuine isomorphism of types.

5.1 Coproduct comparisons

Definition 5.1 (Coproduct extension comparison)**.** The forward and backward maps

$$\mathrm{Ext}(C + D)\, X \;\longrightarrow\; \mathrm{Ext}\, C\, X \;\oplus\; \mathrm{Ext}\, D\, X, \tag{2}$$

$$\mathrm{Ext}\, C\, X \;\oplus\; \mathrm{Ext}\, D\, X \;\longrightarrow\; \mathrm{Ext}(C + D)\, X \tag{3}$$

send a structure whose shape is a left (resp. right) injection to the corresponding summand, and conversely re-inject. The unit comparison $\mathrm{Ext}\, 0\, X \to \emptyset$ is elimination of the empty shape.

Theorem 5.2 (Coproduct comparisons are inverse)**.** *The maps of Definition 5.1 are mutually inverse. In particular*

$$\mathrm{Ext}(C + D)\, X \;\cong\; \mathrm{Ext}\, C\, X \;\oplus\; \mathrm{Ext}\, D\, X, \qquad \mathrm{Ext}\, 0\, X \;\cong\; \emptyset. \tag{4}$$

Proof. By lean. □

5.2 Product comparisons

Definition 5.3 (Product extension comparison)**.** The forward map

$$\mathrm{Ext}(C \times D)\, X \;\longrightarrow\; \mathrm{Ext}\, C\, X \;\times\; \mathrm{Ext}\, D\, X$$

splits a labelling of $P(s) \oplus Q(t)$ along the two injections. The backward map pairs the shapes and combines the labellings by case analysis on the sum of positions.

Theorem 5.4 (Product comparisons are inverse)**.** *The maps of Definition 5.3 are mutually inverse. In particular*

$$\mathrm{Ext}(C \times D)\, X \;\cong\; \mathrm{Ext}\, C\, X \;\times\; \mathrm{Ext}\, D\, X, \qquad \mathrm{Ext}\, 1\, X \;\cong\; 1. \tag{5}$$

Proof. By lean. □

Remark 5.5 (The exponential law)**.** This is the pleasing cross-check. The product container's positions are a *sum*, yet its extension is the pointwise *product* of the extensions. The reason is

$$X^{A+B} \;\cong\; X^A \times X^B : \tag{6}$$

a labelling $P(s) \oplus Q(t) \to X$ *is* a pair of labellings. Currying the sum turns the position-level $\oplus$ into a functor-level $\times$.

5.3 The shared unit y

Definition 5.6 (Identity container as representable)**.** Write y for the container with one shape and one position (also called I when viewed as the unit of $\triangleleft$). Its extension is the identity functor.

Theorem 5.7 (Unit comparison for y)**.** *The maps*

$$\mathrm{Ext}\, y\, X \;\longrightarrow\; X, \qquad X \;\longrightarrow\; \mathrm{Ext}\, y\, X \tag{7}$$

—read the label at the unique position, and place a value at the unique position—are mutually inverse. Thus $\mathrm{Ext}\, y\, X \cong X$*. This is the unit comparison for both* $\otimes$ *and* $\triangleleft$*.*

Proof. By lean. □

6 Discussion

What has been established. The coproduct and product are monoidal structures on **Cont**. The extension functor is strong monoidal for both, via genuine (not identity) comparison isomorphisms. The product/Dirichlet contrast is explained by the exponential law: sum of positions yields product of extensions; product of positions does not split.

Taxonomy of the four structures.

structure	Day convolution of	unit	Ext pointwise?
product $\times$	$(\mathbf{Set}, +, 0)$	$1 = y^0$	yes
Dirichlet $\otimes$	$(\mathbf{Set}, \times, 1)$	$y = y^1$	no
coproduct $+$	not a Day convolution	0	yes
sequential $\triangleleft$	not a Day convolution	$I = y$	no

(The Day-convolution reading of $\times$ and $\otimes$ follows Niu–Spivak; the coproduct fails to preserve the initial object in each variable, and $\triangleleft$ is not cocontinuous in the right variable.)

Machine checking. Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant.

Acknowledgements

The mathematical content follows the theory of containers of Abbott, Altenkirch, and Ghani, and the monoidal taxonomy of polynomial functors. The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.

References

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